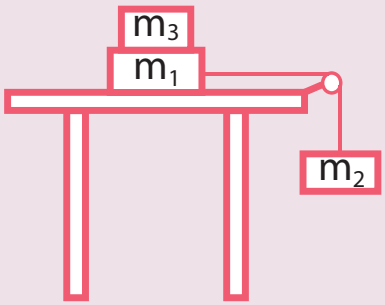
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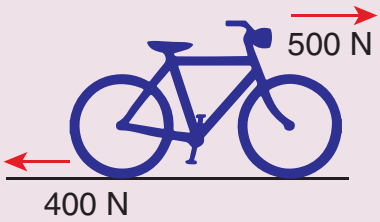
Ans: Tangential acceleration = g sinθ ; centripetal acceleration = $\frac{(T-mg\cos\theta)}{m}$.

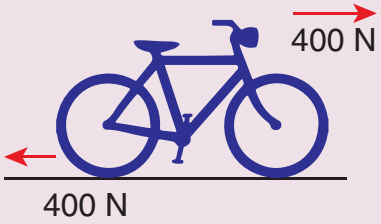
6. Two masses m_1 and m_2 are connected with a string passing over a frictionless pulley fixed at the corner of the table as shown in the figure. The coefficient of static friction of mass m_1 with the table is μ_s. Calculate the minimum mass m_3 that may be placed on m1to prevent it from sliding. Check if m1=15 kg, m2=10 kg, m3=25 and μ_s =0.2



Ans: $m_3 = \frac{m_2}{\mu_s} - m_1$, the combined masses m_1+m_3 will slide.

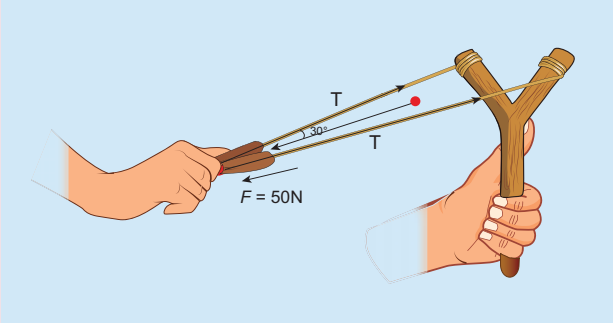
7. Calculate the acceleration of the bicycle of mass 25 kg as shown in Figures 1 and 2.





Ans: a=4 ms^{-2}, zero

8. Apply Lami's theorem on sling shot and calculate the tension in each string ?



Ans: T= 28.868N.

9. A football player kicks a 0.8 kg ball and imparts it a velocity 12 ms^{-1}. The contact between the foot and ball is only for one- sixtieth of a second. Find the average kicking force.

Ans: 576N.

10. A stone of mass 2 kg is attached to a string of length 1 meter. The string can withstand maximum tension 200 N. What is the maximum speed that stone can have during the whirling motion?

Ans: v_{max}=10ms^{-1}

11. Imagine that the gravitational force between Earth and Moon is provided by an invisible string that exists between the Moon and Earth. What is the tension that exists in this invisible string due to Earth's centripetal force? (Mass of the Moon = 7.34×10^{22} kg, Distance between Moon and Earth = 3.84×10^{8} m)

